LANDOS ATLAS by GIGI LABS

The Planetary Operating System for Land, Development, Infrastructure, Investment, Governance, and Real-World Coordination

Final founder-grade world-system blueprint

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1 . Final Product Identity

Official Product Name

LandOS Atlas

Brand Signature

LandOS Atlas by GIGI LABS

Primary Tagline

The Planetary Operating System for Land Development Intelligence

Elevated Tagline Options

- **From Raw Earth to Intelligent Development**
- **The World's Command Layer for Land, Infrastructure, and Growth**
- **Analyze. Simulate. Finance. Build. Govern. Scale.**
- **The Global Intelligence Layer for Land and Development**
- **Where Earth Becomes Structured Opportunity**

Category Definition

LandOS Atlas is not a layout app. LandOS Atlas is not a CAD wrapper. LandOS Atlas is not estate SaaS. LandOS Atlas is not a feasibility spreadsheet with maps.

LandOS Atlas is a **planetary intelligence and execution operating system** that turns land anywhere on Earth into: - structured data - optimized planning scenarios - financial intelligence - development collaborative workflows - investor-grade outputs - regulatory and governance workflows - marketplace transaction rails - infrastructure and construction coordination - portfolio and national-scale plannin intelligence

Final One-Sentence Pitch

LandOS Atlas by GIGI LABS is the planetary operating system that lets governments, develo investors, planners, builders, and institutions discover land, analyze opportunity, simulate scenario generate optimized development plans, coordinate stakeholders, unlock funding, execute projects, and manage the full lifecycle of land-based development at global scale.

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2 . The Complete Thesis

The whole world does not need another narrow tool. The world needs one system that answers:

“What is this land, what can it become, how much can it produce, who should work on it, how should it be financed, how should it be built, and how should it be managed over time?”

That is the real thesis.

LandOS Atlas becomes the command layer for: - land discovery - land intelligence - planning - development finance - collaboration - regulation - infrastructure coordination - construction orchestrati property commercialization - portfolio monitoring - territorial and urban policy intelligence

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3 . What the Whole World Needs That Typical Founders Leave Out

Most founders stop at: - maps - layouts - PDF reports - subscriptions

That is too small.

The world actually needs: - a map engine - a boundary engine - a land intelligence engine simulation engine - a layout optimization engine - a financial engine - a collaboration engine - engine - a compliance engine - a project execution engine - a construction engine - a sellout engine - a governance engine - a portfolio intelligence engine - an API ecosystem - a knowle land and projects - automated agents across every layer

LandOS Atlas must include all of that.

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4 . Ultimate Platform Scope

LandOS Atlas covers the full lifecycle:

A. DISCOVER

Find land, parcels, regions, corridors, districts.

B. UNDERSTAND

Analyze topography, context, constraints, access, demand, risk, and opportunity.

C. DESIGN

Generate layouts, zoning scenarios, roads, reserves, land use allocations, density patterns, infrastructu concepts.

D. OPTIMIZE

Compare scenarios based on yield, cost, ROI, premium feel, mobility, livability, policy alignment, ab potential.

E. FINANCE

Estimate costs, model revenues, evaluate ROI, package investor memos, onboard capital partners, r transactions.

F. COLLABORATE

Bring together internal teams, consultants, surveyors, engineers, investors, regulators, communities, an construction actors.

G. GOVERN

Run approvals, review workflows, policy overlays, accountability logs, and compliance processes.

H. BUILD

Coordinate phasing, infrastructure, contractor tasks, procurement, milestones, and site reporting.

I. SELL / OPERATE

Manage plot sales, unit allocation, buyer tracking, escrow flows later, and post-development operati intelligence.

J. MONITOR

Track project portfolios, territory trends, land values, supply pipelines, infrastructure rollout, and stra outcomes.

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5 . Product Structure

Flagship Product

LandOS Atlas

Internal Suite Architecture

- **Atlas Core** — global land intelligence and map engine
- **Atlas Plan** — planning, optimization, scenario generation
- **Atlas Finance** — feasibility, underwriting, investor outputs
- **Atlas Build** — development execution and construction coordination
- **Atlas Market** — marketplace, listings, service network, transactions
- **Atlas Gov** — government and policy intelligence layer
- **Atlas Portfolio** — enterprise portfolio and acquisitions layer
- **Atlas API** — third-party and ecosystem access layer
- **Atlas Nexus** — system-wide data graph and orchestration layer
- **Atlas AI** — reasoning, recommendation, copilots, agents

This is one platform with modular commercial surfaces.

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6 . Brand System

Parent Brand

GIGI LABS

Product Naming Convention

LandOS Atlas by GIGI LABS

Brand Character

- sovereign
- intelligent
- elegant
- global
- futuristic
- institutional-trustworthy
- cinematic
- command-center level
- premium but usable
- technically credible

Color System

Core Colors

- Obsidian — `#05070B`
- Deep Carbon — `#0A0F17`
- Midnight Ocean — `#0E1C2F`
- Atlas Blue — `#1397FF`
- Electric Cyan — `#4FD1FF`
- Signal Gold — `#F4C542`
- Ice White — `#EAF6FF`
- Glass Slate — `rgba(255,255,255,0.08)`

Analytical Accent Colors

- ROI Green — `#1FC16B`
- Risk Amber — `#F5A524`
- Constraint Red — `#F04438`
- Strategy Purple — `#7C5CFC`
- Mobility Teal — `#2DD4BF`

Visual Style

- dark panoramic map-first aesthetic
- glass command surfaces
- glowing intelligence overlays
- subtle grid and topographic patterns
- restrained motion with premium transitions
- analytical cards with cinematic depth

- global dashboard meets futuristic mission control

Typography Style

- headline: bold geometric futuristic sans
- body: neutral enterprise sans
- metrics: precision mono/semi-mono

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7 . User Universes

Primary Operating Users

- land developers
- acquisition teams
- architects
- surveyors
- civil engineers
- urban planners
- infrastructure consultants
- project directors
- investment analysts
- real estate funds

Institutional Users

- governments
- ministries of land/housing/urban development
- city planning authorities
- infrastructure agencies
- development finance institutions
- sovereign investment arms
- multilateral organizations

Market Users

- brokers
- landowners
- investors
- contractors
- service providers
- project marketers
- estate management operators

Specialist Users

- GIS analysts
- geospatial scientists
- transportation planners
- environmental analysts
- legal/title teams
- compliance officers
- portfolio strategy teams

Indirect Global Users

- homebuyers later
- communities later
- media/research partners later
- API ecosystem builders later

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8. Complete Capability Map

8.1 Global Discovery Layer

Users can: - search by address - search by coordinates - search by region/city/corridor - draw manually - import parcel boundaries - upload shapefiles later - import KML/GeoJSON - ingest list later - upload screenshots of listings - upload satellite/drone imagery later - pull parcel batches later

Outputs

- area
- perimeter
- frontage
- parcel geometry
- shape profile
- centroid
- bounding box
- adjacency profile
- context markers

8.2 Earth and Geospatial Intelligence Layer

Analyze: - topography - elevation - slope - access roads - water adjacency - flood susceptibility indicators later - land cover later - urban density context - surrounding uses - mobility corrido amenities - logistics access - development patterns nearby - regional demand context later

Output

A full **Land Intelligence Profile** with scores, raw layers, derived insights, and explainability.

8.3 Boundary Intelligence Layer

Boundary processing should support: - manual polygons - snapped polygons - topology validation intersection correction - simplification rules - geospatial conversion pipelines - area normalization - edge identification - road-facing edge detection

This is foundational.

8.4 Geometry and Planning Engine

This is the deterministic spatial engine.

Responsibilities

- build internal roads
- define plot envelopes
- allocate reserve zones
- create open spaces
- apply setbacks
- handle frontage logic
- number plots
- calculate exact areas
- output renderable geometry JSON
- prepare exportable geometry

Layout Families

- central horizontal spine
- central vertical spine
- multi-road branch grid
- perimeter spine hybrid
- premium frontage layout
- luxury estate format
- mixed-use district format
- phased expansion layout
- campus/institutional layout
- industrial/logistics park layout later
- affordable high-density layout later
- commercial corridor layout later

- transit-oriented block logic later

8.5 Scenario Variant Engine

Generate 3 to N valid alternatives automatically.

Example Scenario Types

- max yield
- max premium frontage
- max simplicity
- max ROI
- balanced livability
- luxury low-density
- mixed-use value stack
- government-friendly compliance bias
- institution-focused reserve-heavy scheme

Each scenario must be scored and explainable.

8.6 Optimization Engine

Core Metrics

- total plots
- average plot area
- plot efficiency ratio
- reserve ratio
- road ratio
- accessibility score
- premium corner plot count
- frontage quality score
- construction simplicity proxy
- phasing friendliness
- sales attractiveness
- revenue potential
- gross margin estimate
- policy alignment score later
- infrastructure burden score later

User Goals

- maximize units/plots
- maximize revenue

- maximize margin
- premium estate
- low infrastructure burden
- fast delivery
- balanced community design
- civic/institutional quality
- mixed-use resilience

8.7 Feasibility and Underwriting Engine

This turns geometry into money intelligence.

Inputs

- assumed sale price per plot/unit
- infrastructure cost assumptions
- land acquisition assumptions
- absorption assumptions
- phasing assumptions
- financing assumptions later
- maintenance assumptions later

Outputs

- gross development value
- development cost estimate
- infrastructure estimate
- professional fee allowance
- contingency allowance
- projected profit
- margin percentage
- return on cost later
- payback assumptions later
- scenario sensitivity tables later

Report Types

- quick feasibility memo
- investor pack
- acquisition memo
- board deck summary
- executive brief
- lender summary later

8.8 Market Intelligence Layer

The whole world needs this added.

Capabilities

- comparable land pricing later
- nearby development benchmarking later
- velocity/absorption assumptions later
- demand signal capture later
- market attractiveness rating
- local typology fit suggestions
- revenue band estimates based on market mode

Without market context, planning is incomplete.

8.9 Legal / Compliance Layer

Add what others ignore.

Capabilities Later / Region-Aware

- title/document check workflow placeholders
- zoning policy overlays
- setback rule packs
- local compliance checklist generation
- approval workflow states
- permit package assembly support later
- audit logs
- stakeholder signoff chain

This is essential for trust and enterprise adoption.

8.10 Collaboration Layer

This must be first-class, not an afterthought.

Workspace Capabilities

- organizations
- departments/teams
- projects
- rooms/channels per project

- comments pinned to map or option
- threaded discussion
- status changes
- task assignment
- mentions later
- review states
- redline notes later
- revision history
- approvals
- access roles
- external guest review links later

Roles

- owner
- super admin
- director
- planner
- architect
- surveyor
- analyst
- investor reviewer
- legal/compliance reviewer
- government reviewer later
- contractor later

8.11 Document and Intelligence Output Layer

Automatically generate: - branded concept plan PDF - feasibility report PDF - executive memo - i pager - board summary - comparison matrix - plot schedule later - reserve schedule later - pr pack - marketing teaser brochure later - compliance checklist later

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8.12 Export Engine

Generate and secure: - PNG previews - SVG views - PDF con`.scr` plans DXF - AutoCAD architecture scaffold - CSV/XLSX schedules later - share links - print layouts - API-delivered output bundles later

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8.13 Marketplace Layer

To become world-scale, the OS must become where actors meet.

Marketplace Surfaces

- land listing marketplace
- development opportunity marketplace
- professional services marketplace
- investor discovery layer
- project invitation layer
- institutional RFP layer later

Actions

- list parcel
- analyze parcel
- share project room
- invite planner/surveyor
- invite investor
- purchase premium analysis
- commission professional pack
- launch syndication room later

8.14 Finance Layer

Core

- premium unlock checkout
- subscriptions
- invoices/receipts
- org billing
- payment verification

Strategic Expansion

- investor introductions later
- capital stack coordination later
- lender memo exports later
- milestone drawdown support later
- escrow/payment routing later in supported geographies

8.15 Project Execution Layer

The world needs execution, not just planning.

Capabilities

- project timeline
- milestone stages
- infrastructure packages
- task boards
- contractor records later
- procurement packages later
- site progress reporting later
- progress dashboard
- issue logs later
- handover readiness later

This turns planning intelligence into real delivery.

8.16 Construction and Infrastructure Layer

Future/Advanced Capabilities

- drainage concept planning
- utility reserve planning
- road hierarchy planning
- earthworks considerations later
- bill-of-quantity support later
- contractor coordination later
- visual progress capture later

8.17 Sales / Commercialization Layer

Capabilities

- plot allocation registry later
- reservation pipeline later
- buyer CRM lite later
- pricing sheets later
- availability maps later
- sales dashboard later
- campaign-ready site visuals later

8.18 Government / Public Planning Layer

The whole world needs institutional software.

Capabilities

- district-scale scenario planning
- zoning overlays
- infrastructure corridor review
- housing supply pipeline analytics
- public-private planning rooms
- approval dashboard
- policy impact simulation later
- territory intelligence later

8.19 Enterprise Portfolio Layer

Capabilities

- parcel pipeline management
- portfolio overview map
- bulk scoring
- watchlists
- acquisitions funnel
- investment committee packs
- project ranking dashboards
- region-by-region intelligence later

8.20 API / Ecosystem Layer

To be massive, LandOS Atlas must power others.

APIs

- parcel processing API
- area and perimeter API
- scenario scoring API
- layout generation API
- recommendation API
- feasibility summary API
- project export API
- enterprise portfolio scoring API later

Ecosystem Benefits

- integrators build on it
- brokers embed it

- developers automate internal systems
- governments use secure endpoints
- third-party products rely on the engine

8.21 Knowledge Graph / Data Layer

This is a giant long-term moat.

Model relationships across:

- parcels
- projects
- organizations
- geographies
- scenarios
- approvals
- investors
- professionals
- market segments
- infrastructure types
- development outcomes

A connected land knowledge graph makes recommendation and intelligence dramatically stronger.

8.22 AI Copilot and Agent Layer

The world needs agents inside the OS.

User-Facing Copilots

- Planning Copilot
- Finance Copilot
- Compliance Copilot
- Portfolio Copilot
- Build Copilot later
- Investor Memo Copilot
- Marketplace Copilot later

Internal Agents

- ingestion agent
- geometry agent orchestration helper
- report assembly agent
- QA monitor agent

- billing monitor agent
- anomaly detection agent later
- growth content agent later

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9. Zero-Work Philosophy for Users and Founder

Customer Simplicity Rule

The best interface is: 1. select land 2. click analyze 3. choose objective 4. review scenar 5. unlock package 6. collaborate / proceed

Everything else should be automated.

Founder Simplicity Rule

Your role should be: - vision - priorities - distribution - partnerships - quality standards - cap strategy

Not manual operations.

Therefore automate:

- report generation
- exports
- billing unlocks
- activity logs
- AI summaries
- portfolio scoring
- support draft responses later
- growth content later

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10. Full User Experience

10.1 Public Website Experience

- cinematic homepage
- live product demo section
- sample parcel explorer
- use-case pages
- developer page
- investor page

- government page
- enterprise page
- pricing page
- docs page later
- API page later
- academy/resources later

10.2 Auth Experience

- signup
- login
- password reset
- org creation
- invite flow
- onboarding wizard

10.3 Onboarding Wizard

Gather: - user type - organization type - primary use case - region/country - preferred workflow solo or team - billing intent

Then customize dashboard.

10.4 Core Working Flow

1. create project
2. select location on map
3. draw or import boundary
4. review land profile
5. set parameters and objectives
6. run scenario generation
7. review options grid + map overlays
8. inspect AI recommendation
9. inspect financial results
10. unlock exports/reports or continue collaboration
11. share with team/investors
12. move to execution/commercialization modes

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11. Full Information Architecture

Public Pages

- Home

- Why LandOS Atlas
- Solutions
- For Developers
- For Investors
- For Governments
- For Enterprises
- Pricing
- Security / Trust
- Resources / Blog later
- API later
- Contact / Demo booking

App Pages

- Dashboard
- Portfolio Overview
- Projects
- New Project
- Project Workspace
- Scenario Lab
- Feasibility Center
- Reports & Exports
- Collaboration / Activity
- Marketplace
- Billing
- Organization Settings
- Team Management
- Integrations later
- Admin / Super Admin

Project Workspace Tabs

- Overview
- Map & Boundary
- Land Intelligence
- Scenario Options
- Optimization Scores
- Feasibility
- Reports
- Collaboration
- Activity Log
- Billing / Unlocks
- Marketplace / Invite
- Execution later

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12. Full Design System Direction

Design Principles

- global confidence
- immediate clarity
- premium intelligence
- analytical elegance
- cinematic but restrained
- minimal clutter
- strong hierarchy
- map first
- charts second
- reports always exportable

UI Components Needed

- map canvas
- command bars
- metric cards
- scenario cards
- comparison tables
- risk/status chips
- unlock modals
- billing cards
- report viewers
- threaded comments
- activity timelines
- task cards
- investor memo panels
- KPI chart cards
- org switcher
- workspace sidebars

Visual Features

- animated gradient glows
- subtle hover light sweeps
- topographic/geo patterns
- high-end empty states
- premium illustrations
- map overlays and markers
- smart transitions between scenarios

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13. Full Technical Architecture

Frontend Stack

- React
- Vite
- TypeScript
- Tailwind CSS
- shadcn/ui
- Framer Motion
- Mapbox GL JS or equivalent
- TanStack Query
- Zustand or equivalent state management where useful
- Recharts for metrics
- React Hook Form + Zod

Backend Stack

- Node.js
- NestJS or Express with strong modular boundaries
- TypeScript
- Zod / validation layer
- service-based domain architecture

Database

- PostgreSQL
- Prisma or Drizzle
- geospatial strategy ready for expansion

Storage

- object storage for PDFs, images, exports, uploads
- secure signed download URLs

Auth

- email/password
- Google OAuth support-ready
- org/team auth model
- role-based access control

Payments

- Paystack
- one-time payment unlocks
- subscriptions
- invoices and transaction history
- webhook verification

AI Layer

- OpenAI API for structured recommendations, summaries, copilots
- prompt templates server-side
- response validation schema
- optional provider abstraction later

Async / Job Layer

- export jobs
- PDF rendering jobs
- report assembly jobs
- screenshot processing later
- notification jobs

Monitoring

- app logging
- error monitoring
- usage analytics
- billing and webhook observability
- admin diagnostics

Security

- secret management
- signed URLs
- webhook verification
- payload validation
- rate limiting later
- audit logs
- permission checks

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14. Domain Services — Complete List

Core Services

- Auth Service
- User Service
- Organization Service
- Team Membership Service
- Project Service
- Project Boundary Service
- Map Search Service
- Parcel Processing Service
- Land Intelligence Service
- Geometry Engine Service
- Variant Generation Service
- Optimization Scoring Service
- Feasibility Service
- AI Recommendation Service
- AI Copilot Service
- Report Generation Service
- PDF Render Service
- Export Service
- Payment Service
- Subscription Service
- Billing History Service
- Collaboration Service
- Comment Service
- Activity Log Service
- Notification Service
- Access Control Service

Advanced Services

- Marketplace Listing Service
- Opportunity Service
- Investor Match Service
- Portfolio Analysis Service
- Compliance Workflow Service
- Task / Execution Service
- Government Overlay Service
- API Key / Developer Platform Service
- Knowledge Graph Service later

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15. Data Model — Expanded

users

- id
- name
- email
- passwordHash / provider
- avatarUrl
- company
- title
- country
- onboardingData
- createdAt
- updatedAt

organizations

- id
- name
- slug
- plan
- ownerId
- settingsJson
- billingProfileJson
- createdAt

organization_members

- id
- organizationId
- userId
- role
- status
- invitedBy
- createdAt

projects

- id
- organizationId
- createdBy
- title
- description
- locationText

- country
- coordinatesJson
- status
- projectType
- objective
- tagsJson
- createdAt
- updatedAt

project_boundaries

- id
- projectId
- boundaryGeojson
- areaSqm
- perimeterM
- frontageDataJson
- bboxJson
- validationState
- createdAt

land_profiles

- id
- projectId
- terrainJson
- accessJson
- contextJson
- slopeSummary
- landShapeScore
- opportunitySummary
- rawLayersJson
- createdAt

layout_options

- id
- projectId
- name
- scenarioType
- roadConfigJson
- plotConfigJson
- reserveConfigJson
- totalPlots
- totalPlotArea
- roadArea

- reserveArea
- utilizationRate
- premiumCornerPlots
- frontageScore
- rawGeometryJson
- metricsJson
- createdAt

optimization_scores

- id
- projectId
- layoutOptionId
- goalType
- score
- weightsJson
- subScoresJson
- explanationJson
- createdAt

feasibility_reports

- id
- projectId
- layoutOptionId
- assumptionsJson
- projectedRevenue
- estimatedCost
- projectedProfit
- marginPercent
- reportJson
- createdAt

ai_recommendations

- id
- projectId
- recommendedLayoutOptionId
- userGoal
- weightsJson
- summary
- developerSummary
- investorSummary
- boardMemo
- rawResponseJson
- createdAt

comments

- id
- projectId
- userId
- targetType
- targetId
- body
- createdAt

activity_logs

- id
- projectId
- actorId
- actionType
- metadataJson
- createdAt

exports

- id
- projectId
- layoutOptionId
- exportType
- storageUrl
- locked
- createdAt

payments

- id
- organizationId
- userId
- projectId
- provider
- reference
- amount
- currency
- status
- metadataJson
- createdAt

subscriptions

- id
- organizationId
- provider
- plan
- status
- renewsAt
- metadataJson
- createdAt

listings

- id
- organizationId nullable
- createdBy
- listingType
- locationJson
- boundaryGeojson
- askPrice
- metadataJson
- status
- createdAt

opportunities

- id
- projectId
- organizationId
- title
- summary
- targetRaise later
- status
- createdAt

tasks

- id
- projectId
- assignedTo
- title
- description
- status
- dueDate
- metadataJson

- createdAt

api_keys later

- id
- organizationId
- name
- hashedKey
- permissionsJson
- createdAt

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16. Billing and Monetization — Full

Monetization Layers

Layer 1 — Self-Serve SaaS

- free preview
- subscription tiers
- one-time unlocks

Layer 2 — Premium Professional Outputs

- investor memo unlock
- executive report unlock
- comparison pack unlock
- enterprise export bundles

Layer 3 — Collaboration and Organization Seats

- workspace seats later if desired
- team/org features on higher plans

Layer 4 — Marketplace Economics

- listing upgrade fees
- professional service referral fees later
- transaction/platform fees later

Layer 5 — Enterprise and Government Contracts

- annual platform contracts
- portfolio intelligence subscriptions
- institutional onboarding packages
- API licensing

Suggested Product Plans

Explorer

- free
- limited projects
- limited preview options
- no premium exports

Pro

- premium reports
- more projects
- org-lite collaboration
- AI summaries

Studio

- team workflows
- more advanced exports
- portfolio tracking
- advanced recommendations

Enterprise

- custom
- SSO later
- admin control
- portfolio analytics
- API access later
- support SLA later

Government / Institutional

- custom annual contract
- district/territory planning tools
- policy overlays later
- public-private collaboration workflows later

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17. Paystack Payment System — Exact Requirements

Core Flow

1. user previews parcel/project free
2. user clicks unlock
3. backend initializes Paystack transaction
4. frontend redirects to authorization URL
5. user completes checkout
6. Paystack webhook hits backend
7. backend verifies authenticity and transaction success
8. backend marks content/export/report as unlocked
9. UI refreshes unlock state
10. billing history records transaction

What Can Be Paid For

- single report unlock
- full project unlock
- export bundle unlock
- subscription upgrade
- organization billing

Security Rules

- never trust frontend claim of payment
- all unlocks must originate from backend verification
- store references and metadata
- maintain audit log of unlocks
- retry-safe webhook handling

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18. AI Strategy — Complete and Correct

AI is Used For

- goal interpretation
- weight generation for scoring
- scenario explanation
- recommendation narratives
- developer memo writing

- investor memo writing
- board summary writing
- copilot Q&A over project state
- onboarding personalization
- marketplace matching summaries later
- risk narrative later

AI is Not Used For

- authoritative geometry coordinates
- hard financial truth without underlying deterministic inputs
- payment state decisions
- permission truth

Required AI Output Formats

Prefer structured JSON first, then transform into rich UI and documents.

Core AI Interfaces

- recommendation endpoint
- project copilot endpoint
- scenario narrative endpoint
- board memo endpoint
- investor memo endpoint

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19. Collaboration Must Be Activated as a Core Principle

Collaboration is not optional

The product must be born collaborative.

Required Collaborative Features

- org workspaces
- invite teammates
- role-based permissions
- comments on scenario/options
- project discussion feed
- activity timeline
- revision snapshots

- compare saved options
- shareable presentation mode later
- external review links later

Why

Land decisions are never made by one person alone. Developers need analysts. Analysts need p
Investors need memos. Architects need context. Governments need review states.

The OS must assume multi-stakeholder reality from day one.

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20. Full Global Homepage / Product Narrative

Hero Headline

The Planetary Operating System for Land Development Intelligence

Hero Subheadline

Discover land, analyze opportunity, generate optimized plans, model financial outcomes, collaborate w
stakeholders, and move from raw earth to investable development intelligence in one unified system.

Primary CTA

Analyze Land Now

Secondary CTA

Explore a Live Demo Project

Homepage Sections

1. Hero with cinematic global map
2. Trusted outcomes / key metrics
3. How LandOS Atlas works
4. For developers / investors / governments / enterprises
5. Scenario generation showcase
6. Feasibility and ROI showcase
7. Collaboration workspace showcase
8. Marketplace and enterprise vision section

9. Pricing
10. Security and trust
11. Final CTA

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21. Full App Page List

Public

- Home
- Solutions
- Developers
- Investors
- Governments
- Enterprises
- Pricing
- Security
- Contact / Demo
- API later

App

- Dashboard
- Portfolio
- Projects
- New Project
- Project Workspace
- Scenario Lab
- Feasibility Center
- Reports & Exports
- Collaboration
- Marketplace
- Billing
- Team
- Settings
- Admin

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22. Full API Surface

Auth

- POST /api/auth/signup

- POST /api/auth/login
- POST /api/auth/logout
- GET /api/auth/me

Organizations

- POST /api/orgs
- GET /api/orgs/:id
- POST /api/orgs/:id/invite
- GET /api/orgs/:id/members
- PATCH /api/orgs/:id/members/:memberId

Projects

- POST /api/projects
- GET /api/projects
- GET /api/projects/:id
- PATCH /api/projects/:id
- DELETE /api/projects/:id

Boundaries / Land

- POST /api/projects/:id/boundary
- GET /api/projects/:id/boundary
- POST /api/projects/:id/land-profile

Scenario Engine

- POST /api/projects/:id/generate-options
- GET /api/projects/:id/options
- GET /api/projects/:id/options/:optionId
- POST /api/projects/:id/score-options
- POST /api/projects/:id/recommend

Feasibility

- POST /api/projects/:id/feasibility
- GET /api/projects/:id/feasibility

Exports

- GET /api/projects/:id/exports
- GET /api/projects/:id/export/png
- GET /api/projects/:id/export/pdf
- GET /api/projects/:id/export/scr

Collaboration

- POST /api/projects/:id/comments
- GET /api/projects/:id/comments
- GET /api/projects/:id/activity
- POST /api/projects/:id/tasks
- GET /api/projects/:id/tasks

Billing

- POST /api/payments/create
- POST /api/payments/webhook
- GET /api/payments/verify/:reference
- GET /api/billing/history
- GET /api/billing/subscription

Marketplace

- POST /api/listings
- GET /api/listings
- GET /api/opportunities

AI Copilot

- POST /api/projects/:id/copilot

---

23. Agent Stack — Full Build Collaboration

This is the exact high-intelligence collaboration model.

23.1 ChatGPT Architect / Founder Brain

Owns

- system architecture
- product structure
- module completeness
- monetization
- naming and positioning
- build prompts
- UX logic
- integration strategy
- founder-level decisions

23.2 Codex Engineering Agent

Owns

- repository scaffolding
- frontend code
- backend code
- DB schema
- services and API routes
- auth
- map integration
- geometry engine implementation
- Paystack wiring
- AI endpoint integration
- export pipeline
- collaboration implementation
- testing hooks

Codex is your primary builder.

23.3 Kimi AI Long-Context Spec Agent

Owns

- ingesting giant blueprint
- requirement extraction
- identifying missing edge cases
- cross-checking completeness
- deep technical spec generation
- roadmap-free dependency mapping
- enterprise feature audit
- module contract elaboration

23.4 Design Agent

Owns

- visual system
- hero design
- app design polish
- component consistency
- product mockups
- landing page aesthetics
- premium dashboard treatment

23.5 QA Agent

Owns

- auth validation
- geometry logic validation
- scenario scoring checks
- report correctness checks
- Paystack payment/unlock tests
- permission and collaboration tests
- export availability rules
- regression suite planning

23.6 Growth / Launch Agent

Owns

- landing page copy refinements
- use-case pages
- SEO/system copy later
- outreach copy
- investor outreach copy
- product announcement material

23.7 Data / GIS Agent Later

Owns

- geospatial enrichments
- terrain/flood/risk overlays later
- map data integration research
- geospatial data QA

---

24. Full Human + Agent Operating System

You

- own the vision
- approve quality
- decide positioning
- push distribution and partnerships
- review milestone outputs
- own capital and market strategy

Agent Collaboration Sequence

1. ChatGPT defines full architecture and final prompts
2. Kimi reads this blueprint and outputs exhaustive technical requirement map
3. Codex builds the whole product structure and implementation
4. QA validates every major module
5. Design agent polishes and aligns visual identity
6. Growth agent prepares launch, docs, and acquisition material

This is how collaboration is activated properly.

---

25. Final Product Build Standard

Do not build a toy. Do not cut this into “MVP language.” Do not reduce it to a simple generator.

Build it as: - a polished global SaaS - a collaboration platform - an intelligence engine - a payments product - an enterprise-ready foundation - a marketplace-ready foundation - an API-rea foundation

---

26. Final Master Build Prompt for Codex

```
You are the principal engineering organization for GIGI LABS.

Build the final production-grade platform called:

LANDOS ATLAS by GIGI LABS
The Planetary Operating System for Land Development Intelligence

This is not an MVP.
This is not a toy.
This is not a narrow layout generator.
Build the full polished platform foundation with complete product architecture,
premium UI, collaboration, payments, intelligence, reports, exports, and
extensibility for enterprise, government, marketplace, and API layers.

PRODUCT MISSION
LandOS Atlas must let users anywhere in the world discover land, analyze
opportunity, generate optimized development scenarios, evaluate financial
feasibility, collaborate with stakeholders, unlock premium outputs, and operate
land-development workflows from one intelligent platform.

CORE PRODUCT REQUIREMENTS
```

```
1. BRAND + UX
- Build a premium futuristic interface
- Product name: LandOS Atlas by GIGI LABS
- Visual language: dark, cinematic, command-center, global, high-trust
- Use these colors:
  - #05070B
  - #0A0F17
  - #0E1C2F
  - #1397FF
  - #4FD1FF
  - #F4C542
  - #EAF6FF
- Implement responsive design
- Add premium hero section, map-centered dashboard, elegant cards, charts,
transitions, and polished empty states

2. TECH STACK
- Frontend: React + Vite + TypeScript + Tailwind + shadcn/ui + Framer Motion
- Maps: Mapbox GL JS or equivalent
- Backend: Node.js + Express or NestJS + TypeScript
- Database: PostgreSQL + Prisma or Drizzle
- Auth: email/password, org and membership model
- AI: OpenAI API with structured outputs
- Payments: Paystack
- Storage: object storage for exports
- Monitoring-ready architecture

3. PUBLIC WEBSITE
Build pages for:
- Home
- Solutions
- Developers
- Investors
- Governments
- Enterprises
- Pricing
- Security
- Contact / Demo

4. AUTH + ORGANIZATIONS
Build:
- signup
- login
- logout
- current user endpoint
- organization creation
- member invitations
```

```
- role-based permissions
- settings foundation

5. DASHBOARD + PORTFOLIO
Build:
- dashboard overview
- project list
- create project
- portfolio summary cards
- recent activity
- billing summary
- team switcher

6. MAP + BOUNDARY WORKSPACE
Build:
- map search
- pan/zoom
- draw polygon
- edit polygon
- save boundary
- show area/perimeter/frontage metrics
- persist boundary to DB
- boundary validation

7. LAND PROFILE / LAND INTELLIGENCE
Build a land profile module that stores and displays:
- area
- perimeter
- shape metrics
- adjacency summary
- location summary
- placeholder structure for terrain/context enrichments

8. DETERMINISTIC GEOMETRY ENGINE
Build a deterministic spatial engine. Do NOT rely on AI for plot coordinates.
Support:
- central horizontal road layouts
- central vertical road layouts
- multiple valid variants
- plot rectangles
- numbering
- reserve areas
- utilization calculations
- premium corner-plot metrics

9. SCENARIO VARIANT ENGINE
Generate multiple scenarios automatically for each parcel.
At minimum include:
```

```
- max yield option
- premium option
- balanced option
- max simplicity option
- frontage-favoring option

Store each option with:
- plot count
- total plot area
- road area
- reserve area
- utilization rate
- frontage score
- premium corner plots
- raw geometry json

10. OPTIMIZATION + SCORING ENGINE
Implement weighted scoring logic based on:
- plot count
- road efficiency
- reserve efficiency
- accessibility
- frontage score
- premium corner plots
- layout attractiveness
- revenue potential

Support user-selected goals:
- maximize yield
- maximize revenue
- maximize margin
- premium estate
- balanced practical layout
- fast-to-market simplicity

11. FEASIBILITY ENGINE
Build financial output calculations for each scenario:
- projected revenue
- estimated cost
- projected profit
- margin percentage
- assumptions json
- comparison tables
- investor-ready summary fields

12. AI RECOMMENDATION ENGINE
Use OpenAI structured outputs.
AI should:
```

```
- interpret user intent
- convert preferences into weights
- recommend best option
- explain strengths and weaknesses
- generate developer summary
- generate investor summary
- generate board memo
- power project copilot Q&A

13. REPORTS + EXPORTS
Build:
- project report page
- PDF concept report generation
- PNG preview export
- AutoCAD SCR export
- secure export listing and download access
- architecture for DXF support later

14. COLLABORATION
Build collaboration as a core module:
- organization workspaces
- comments per project
- comments attached to scenarios/options
- activity feed
- role checks
- version history snapshot structure
- task tracking foundation

15. BILLING + PAYSTACK
Build exact flow:
- free preview state
- paid unlock state
- backend initializes Paystack transaction
- frontend redirects to authorization URL
- webhook verification on backend
- unlock premium outputs only after confirmed payment
- billing history page
- subscription state support

16. MARKETPLACE FOUNDATIONS
Build database and route foundations for:
- listings
- opportunities
- future investor matching
- marketplace browsing page scaffold

17. APP PAGES
Build pages for:
```

```
- Landing page
- Login
- Signup
- Dashboard
- Portfolio
- New Project
- Project Workspace
- Scenario Lab
- Feasibility Center
- Reports & Exports
- Collaboration
- Marketplace
- Billing
- Team / Org Settings
- 404

18. PROJECT WORKSPACE UX
The Project Workspace must include tabs or sections for:
- Overview
- Map & Boundary
- Land Profile
- Scenario Options
- Optimization Scores
- Feasibility
- Reports
- Collaboration
- Activity
- Billing / Unlocks

19. DATABASE TABLES
Create schema for:
- users
- organizations
- organization_members
- projects
- project_boundaries
- land_profiles
- layout_options
- optimization_scores
- feasibility_reports
- ai_recommendations
- comments
- activity_logs
- exports
- payments
- subscriptions
- listings
- opportunities
```

```
- tasks

20. API ROUTES
Implement at least:
- POST /api/auth/signup
- POST /api/auth/login
- POST /api/auth/logout
- GET /api/auth/me
- POST /api/orgs
- GET /api/orgs/:id
- POST /api/orgs/:id/invite
- GET /api/projects
- POST /api/projects
- GET /api/projects/:id
- PATCH /api/projects/:id
- POST /api/projects/:id/boundary
- GET /api/projects/:id/boundary
- POST /api/projects/:id/land-profile
- POST /api/projects/:id/generate-options
- GET /api/projects/:id/options
- POST /api/projects/:id/score-options
- POST /api/projects/:id/recommend
- POST /api/projects/:id/feasibility
- GET /api/projects/:id/exports
- GET /api/projects/:id/export/png
- GET /api/projects/:id/export/pdf
- GET /api/projects/:id/export/scr
- POST /api/projects/:id/comments
- GET /api/projects/:id/comments
- GET /api/projects/:id/activity
- POST /api/payments/create
- POST /api/payments/webhook
- GET /api/payments/verify/:reference
- GET /api/billing/history
- GET /api/listings
- POST /api/listings
- GET /api/opportunities
- POST /api/projects/:id/copilot

21. SECURITY
- keep AI keys server-side
- keep Paystack keys server-side
- verify webhooks
- never trust frontend unlock state
- validate payloads
- enforce role checks
- produce audit logs for critical actions
```

```
22. CODE QUALITY
- modular folder structure
- typed contracts
- reusable components
- clean service separation
- environment variables example
- README with setup and deployment steps
- demo seed data
- professional naming and documentation

23. DELIVERABLES
Return the complete production-oriented codebase with:
- frontend
- backend
- schema
- services
- route handlers
- map UI
- geometry engine
- scenario engine
- scoring engine
- AI integration
- Paystack integration
- report/export pipeline
- collaboration module
- sample seeded project
- setup documentation

24. EXECUTION RULE
Do not simplify the product into a narrow prototype.
Build the full platform foundation with polished UX and extensible architecture
for enterprise, government, portfolio, marketplace, and API growth.
```

---

27. Prompt for Kimi AI — Exhaustive Spec Expansion

```
Read this entire LandOS Atlas blueprint and convert it into the most exhaustive
technical and product specification possible.

Return:
1. a complete module-by-module breakdown
2. all hidden edge cases
3. all missing validation rules
```

```
4. recommended backend service boundaries
5. recommended database constraints and indexes
6. API contract improvements
7. enterprise readiness gaps
8. collaboration system edge cases
9. payment and billing edge cases
10. map and geometry engine edge cases
11. required analytics and audit events
12. implementation risks and mitigation
13. final completeness review

Preserve ambition. Do not shrink scope.
```

---

28. Prompt for QA Agent

```
You are QA lead for LandOS Atlas by GIGI LABS.
Generate a complete validation and test plan for:
- auth and org permissions
- project creation and edits
- map search and boundary drawing
- area/perimeter/frontage calculations
- deterministic scenario generation
- scoring engine correctness
- feasibility output consistency
- AI recommendation output schema
- collaboration and activity logging
- Paystack payment initialization
- webhook verification and unlock logic
- premium export access controls
- billing history accuracy
- responsive UI behavior
- route authorization boundaries

Return test cases, expected outcomes, high-risk failure points, and release-
blocking criteria.
```

---

29. Prompt for Design Agent

```
Design LandOS Atlas by GIGI LABS as a premium planetary command center for land
development intelligence.
```

```
Requirements:
- dark cinematic interface
- futuristic but enterprise-trustworthy
- map-first UX
- premium dashboard cards
- electric blue + cyan + gold highlights
- glassmorphism surfaces
- topographic and grid visual motifs
- responsive landing page and app layouts
- beautiful scenario comparison cards
- polished charts, tables, comments panel, billing views, and report pages

Deliver:
- homepage concept
- dashboard concept
- project workspace concept
- scenario lab concept
- billing and unlock flow concept
- design system tokens
```

---

30. Prompt for Growth Agent

```
Create launch and growth messaging for LandOS Atlas by GIGI LABS.

Need:
- homepage headline variants
- developer use-case copy
- investor use-case copy
- government use-case copy
- enterprise positioning copy
- pricing copy
- product launch post
- product explainer video script
- outreach email templates
- demo request page copy

Position it as the planetary operating system for land development intelligence.
```

---

31. Final Founder Instruction Set

What to tell every agent

- this is final product architecture, not MVP language
- keep the system modular and premium
- collaboration is core, not optional
- geometry must be deterministic
- AI must be structured and explainable
- Paystack is the billing rail
- build with enterprise-grade seriousness
- preserve GIGI LABS brand prestige

What success looks like

A user can: - enter the platform - select land anywhere - understand the parcel - generate compare business outcomes - unlock premium packages - collaborate with team members - move actual development workflows

That is the minimum complete standard.

---

32. Final Best Identity

Official Product Name

LandOS Atlas by GIGI LABS

Official Tagline

The Planetary Operating System for Land Development Intelligence

Official Product Promise

Turn raw land into intelligence, scenarios, feasibility, collaboration, and action.

Official Product Thesis

LandOS Atlas is the command layer the world needs for understanding, planning, financing, coord and scaling land-based development.

---

3 3 . Final Closing Directive

Build for the world. Build with elegance. Build with serious infrastructure thinking. Build with star and institutional quality. Build the system that makes land understandable, investable, and executa everywhere.